MILITARY SPECIFICATION

TEST SET, DOPPLER RADAR AN/APM-148(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for flight line use and shall provide for testing and evaluation of the Navigation Set, Radar AN/APN-130(V) while the navigation set is installed in its normal operating configuration in an aircraft.

1.2 Classification - The Doppler Radar Test Set shall consist of the following items:

Items	Type Designation	Appl. Para.
Simulator, Doppler Signal	SM-229(*)/APM-148	3.4.1
Control-Indicator	C-3667(*)/APM-148	3.4.2
Cable Assembly	W1 of AN/APM-148	3.4.3
Cable Assembly	W2 of AN/APM-148	3.4.4
Cable Assembly	W3 of AN/APM-148	3.4.5
Cable Assembly	W4 of AN/APM-148	3.4.6
Cable Assembly	W5 of AN/APM-148	3.4.7
Truck, Hand, Platform		3.4.8

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5086	Wire, Electrical, 600 Volt, Copper, Aircraft
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

FSC 6625

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-R-6855	Rubber; Synthetic, Sheet, Molded and Extruded, for Aircraft Applications
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MS 21993	Cable Assemblies, Power, Electrical
MS 28053	Meter, Time Totalizing, 115 Volt, 400 Cycle

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

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(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The test equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the test equipment, including cables and platform hand truck, shall be a minimum consistent with good design and shall not exceed 538 pounds.

3.2.3 Reliability

3.2.3.1 Operational Stability - The test equipment shall operate with satisfactory performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.3.2 Operating Life - The test equipment shall have a minimum total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 100 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.3.4 Time Totalizing Meter - The doppler signal simulator shall contain a time totalizing meter in accordance with Specification MIL-M-7793 and MS 28053.

3.2.4 Cabling and Connections

3.2.4.1 Cables and Connectors - The test equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.4.2 Input Power Connection - Input power cable and connections shall be in accordance with MS 21993.

3.2.5 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-T-21200.

3.2.6 Interchangeability - The test equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

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3.2.7 Interference Control - The generation of radio interference by the test equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for test equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC

3.2.11 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 2 equipment, except as modified herein.

3.2.11.1 Vibration - Specified performance shall be tested under standard conditions only.

3.2.12 Warm-Up Time - The time required for the test equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 20 minutes under standard conditions and 40 minutes at extreme service conditions.

3.2.13 Primary Input Power Requirements - The test equipment shall meet all applicable requirements of Specification MIL-T-21200 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-T-21200 having limits as specified herein. The power required shall not exceed the specified amount.

(1) AC Power (Single Phase) 115 V, 50-70 cps and 380-420 cps, 490 VA.

3.2.13.1 Undervoltage Protection - The test equipment shall not be damaged when exposed to transient voltages as defined in Figure 2 of MIL-STD-704, and shall automatically resume normal operation when the voltage returns within limits of normal operation.

3.2.14 Standby Provisions - The test equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Frequency Characteristics - The test set shall provide for generation of simulated doppler returns of 103.4, 990, and 1980 cps, ± 5% under standard conditions. Frequency drift shall not exceed + 1% or -1-1/2% of the frequency obtained under standard conditions, and shall not exceed ± 2% from -40°C to +20°C and +40°C to +71°C.

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3.3.3 Modulation Characteristics - The test set shall accept a continuous-wave RF input of 13.3 ± 0.2 kmc and shall provide a nominal output of 13.3 kmc on which is imposed simulated doppler returns of 103.4, 990, and 1980 cps ± 5% under standard conditions.

3.3.4 Sensitivity - The test set nominal output power shall be 111 db below the input signal power level.

3.3.5 Functional Characteristics - The doppler radar test set shall provide for reception of microwave energy radiated by Navigation Set, Radar AN/APN-130(V), for confinement of this radiated energy within the test set and for suppression of input from other external sources. The test set shall also provide for the development of simulated doppler signals of 103.4, 990 and 1980 cps ± 5% under standard conditions and the return of these doppler signals of the RF carrier to the receiving section of the Radar Navigation Set antenna, thus generating a simulated doppler signal which simulates aircraft directional groundspeed velocities for operation of the Radar Navigation Set. Remote control facilities shall be incorporated in the test set for operators use during performance checkout of the navigation system. The test set shall conform physically to the structural outline of a receiver-transmitter. A mounting configuration shall be provided to meet this requirement.

3.4 Detail Requirements

3.4.1 Simulator, Doppler Signal SM-229/APM-148 - Simulator, Doppler Signal SM-229/APM-148 shall be used on Receiver-Transmitter, Radar RT-591/APN-130. The doppler signal simulator shall meet the following requirements:

3.4.1.1 Function - The doppler signal simulator shall provide for reception of microwave radiation from Receiver-Transmitter, Radar RT-591/APN-130 and from the return of microwave energy, which is attenuated and which has simulated modulation signals imposed upon the microwave carrier, to the receiver section of the receiver-transmitter. The doppler signal simulator shall provide operating power (by means of integral power supplies) for relays and motor operation within the unit and shall also supply power for amplifier and indicator circuitry within the control-indicator.

3.4.1.2 Form Factor - The dimensions of the doppler signal simulator shall not exceed 59 inches in length by 33 inches in width by 29 inches in height.

3.4.1.3 Weight - The weight of the doppler signal simulator shall not exceed 250 pounds.

3.4.1.4 Contents - The doppler signal simulator shall consist of the following assemblies, subassemblies, and circuits:

(1) Enclosure Assembly - The enclosure assembly shall provide for housing and mounting of the component parts of the doppler signal simulator. The enclosure shall provide an operation panel for control purposes and electrical connections for inter-connection to external circuits. The enclosure shall also provide absorbent material for shielding undesired received power input and extraneous outputs. The configuration of the enclosure shall be such as to meet the specific installation requirements of the aircraft in which the Navigation Set, Radar AN/APN-130(V) is installed.

(2) Motor Assembly - The motor assembly shall consist of a DC powered drive motor, electro-mechanical clutches, gearbox assemblies, shafts, and serrated reflecting wheels. The motor assembly shall provide drive for the serrated reflecting wheels through the electro-mechanical linkage.

(3) R-F Section - The R-F section of the doppler signal simulator shall consist of waveguide assemblies, an R-F attenuator, and R-F switching. The waveguide assemblies shall provide for reception of R-F energy in the Ku

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band from the antenna reflectors in the doppler signal simulator, routing of R-F energy to the simulator wheel cavities through the R-F attenuator and switches, routing of R-F energy from the simulator wheel cavities, and transmission of R-F energy to the antenna reflectors. The R-F attenuator shall be a ganged attenuator, manually operated from a common shaft having four separate equal attenuator pads providing attenuation through the range 1 to 40 db, controlling the power level supplied to the receiver portion of the Receiver-Transmitter, Radar RT-591/APN-130.

(4) Power Supplies - Unregulated and shunt regulated power supplies shall be provided. The shunt regulated power supplies shall provide power outputs as follows: +22 to 30 V DC at 1.25 amperes with a ripple content not to exceed 3.0 V RMS for the operation of indicator lamps and relays in the control-indicator unit, and relays in the Doppler Signal Simulator, 24 ± 2 V DC at 0.7 ampere with a ripple content not to exceed 1.5 V RMS for the operation of the motor, clutches, and relays in the doppler signal simulator. The unregulated supply shall provide +55 (-10 +15) V DC at 150 ma with a ripple content not to exceed 0.5 V RMS for the operation of series regulated power supply and an amplifier in the control-indicator.

3.4.1.5 Controls and Indicators - The following operational controls and indicators shall be located on the doppler signal simulator control panel:

Name	Positions	
DB ATTENUATION Control	Continuous 1 to 40 db Indication	Adjusts R-F Attenuation with an accuracy of ± 0.5 db or ± 2%, whichever is greater, throughout the range of 1-30 db. From 30-40 db, the accuracy shall be within 2 db.
COUNT ON switch	On - Off (Toggle)	Operates the Nautical Miles Indicator (M1)
NAUTICAL MILES Indicator M1	Counter indicator (4 digit)	Provides indication of nautical miles traveled based upon an integration of simulated cycles. The counter reading will be accurate to within 1% of the simulated input provided the miles traveled exceeds one mile.
CB1 Circuit Breaker	Open Closed	Provides circuit overload protection.
CB2 Circuit Breaker	Open Closed	Provides circuit overload protection.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	MS3102A-10SL-3P	Provides primary power inlet
J2	Bendix Pigmy SP02-22-215(W) (or equivalent)	Provides power outlet to the Control-Indicator C-3667/APM-148

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3.4.2 Control-Indicator C-3667/APM-148 - The control-indicator shall meet the following requirements:

3.4.2.1 Function - The control-indicator shall permit remote operational control of the doppler radar test set for evaluation and display of the navigation set operating condition. Signals and voltages, indicating the operating condition of the navigation set, shall be displayed by meters, and go, no-go indicator lamps on the control-indicator panel.

3.4.2.2 Form Factor - The dimensions of the control-indicator shall not exceed 20 inches in length by 13 inches in width by 11 inches in height. The control-indicator shall be enclosed in a combination case in accordance with Specification MIL-T-21200, except that the case may provide for external connection to be made through the rear as a means of facilitating its use in the aircraft.

3.4.2.3 Weight - The weight of the control-indicator shall not exceed 40 pounds.

3.4.2.4 Contents - The control-indicator shall consist of the following assemblies, subassemblies and circuits:

(1) Test Meter - The test meter shall provide for visual display of output voltage levels from the Power Supply PP-2712/APN-130. The meter shall be calibrated from 18 to 32 V DC. The DC scale shall be red. The meter shall also provide an AC scale calibrated from 105 to 125 V RMS. The meter shall operate over the frequency range of 50 to 450 cps. The AC scale shall be black. The test meter shall not exceed 3-1/2 inches in diameter and 4-1/2 inches in length. The weight of the test meter shall not exceed 2-1/2 pounds.

(2) Preamplifier Signal-to-Noise Ratio Meter - The preamplifier signal-to-noise ratio meter shall provide for visual display of the audio amplifier signal output level from the Receiver-Transmitter, Radar RT-591/APN-130. The meter shall be calibrated in db. The preamplifier signal-to-noise ratio meter shall not exceed 3-1/2 inches in diameter by 4-1/2 inches in length. The weight of the preamplifier signal-to-noise ratio meter shall not exceed two pounds.

(3) Lamps - Lamps shall be provided as follows:

Indication	Color	Function
TEST SET POWER	White	When illuminated, indicates that primary power is being applied to the test set.
TEST SET READY	Green	When illuminated, indicates that the test set is connected to the navigation set test connector 4J4 on the Power Supply PP-2712/APN-130 and that the high voltage is applied to the klystron.
AC COMPUTER GO NO-GO	Green Amber	Indicates whether AC velocity output voltages are within acceptable limits. Limits shall be as follows: VH and VD 6% ± 1%; tolerance shall be an additional ± 1% from -40°C to +71°C. VZ ±15% ± 1%; tolerance shall be an additional ± 3% from -40°C to +71°C.

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Indication	Color	Function
20 V DC GO NO-GO	Green	When illuminated indicates that power supply ± 20 V DC levels are in tolerance. Tolerances are as follows: Upper limit 20.2 ± .051 -0.0 volts and lower limit 19.8 + 0.0 -0.05 volts from -40°C to +71°C.
B CLIPPER + -	White White	When illuminated indicates voltage polarity from B Clipper.

3.4.2.5 Controls - The following operational controls shall be located on the front panel of the control-indicator:

	Name	Positions	Function
(1)	POWER SUPPLY	+20 VDC -20 VDC +28 VDC +26.5 VDC 115V 400 CPS PH B 115 V EXC	Selects Navigation Set, Radar AN/APN-130(V) voltages for presentation on the test meter.
(2)	DOPPLER FREQ (CPS)	A (FWD) B (FWD) C (FWD) OFF D (REV) E (REV) F (REV)	Energizes the test set and selects the motor speed and direction.
(3)	"B" CLIPPER OUT	FWD PORT FWD STBD AFT PORT	Selects Navigation Set, Radar AN/APN-130(V) voltages for presentation on the test meter and selects B clipper test parameters.
(4)	SDC SIGNAL	PORT STBD AFT B FWD A AFT A FWD B FWD B AFT A FWD A AFT B	Selects Navigation Set, Radar AN/APN-130(V) voltages for presentation on the test meter and selects the audio amplifier output test parameters.
(5)	VELOCITY AC	CONF TEST DRIFT HEADING VERT	Selects Navigation Set, Radar AN/APN-130(V) voltages for presentation on the test meter and establishes predicted simulated aircraft track and provides self check of the test circuitry.
(6)	FILTER SELECT	400 CPS 800 CPS 990 CPS SDC INPUT	Selects the desired band pass filter for confidence check of Navigation Set, Radar AN/APN-130(V) crystals and preamplifiers.
(7)	REFERENCE ADJUST	270° rotation	Sets a reference on the preamplifier signal-to-noise ratio meter.

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3.4.2.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type (or Equivalent)	Function
2J1	Bendix Pigmy SP02-20-16PW	Parameter inlet for Control Indicator.
2J2	Bendix Pigmy SP02-18-11P	Parameter inlet for Control Indicator.
2J3	Bendix Pigmy SP02-20-16P	Supplies Converter, Signal Data CV-971/APN-130 test parameter inlet from Control-Indicator.
2J4	Bendix Pigmy SP02-22-21P	Supplies power and control inlet from Doppler Signal Simulator.
1J1	MS-3102-10SL-3P	Supplies primary power inlet to the Doppler Signal Simulator.
1J2	Bendix Pigmy SP02-22-21SW	Supplies power inlet to the Control Indicator from the Doppler Signal Simulator.

3.4.2.7 Circuits

(1) Regulated Power Supply Circuit - The regulated power supply shall provide a power output of 28 ± 0.25 V DC at 75 ma with a ripple content not to exceed 50 MV RMS for the operation of amplifiers and an electrical relay circuit in the control-indicator.

(2) Difference Circuit - The difference circuit shall provide a means of sampling the system AC computer output voltages V_h, V_d, and V_z.

(3) Difference Circuit Amplifiers - Difference circuit amplifiers shall be provided for amplification of the difference circuit AC output voltages.

(4) Electrical Relay Readout Circuit - The electrical relay readout circuit shall provide DC voltage outputs to indicator lamps.

(5) SDC Input Preamplifier Circuit - The SDC input preamplifier circuit shall receive input signals from the Signal Data Converter CV-971/APN-130 and shall amplify these signals for visual display on the preamplifier signal-to-noise ratio meter.

(6) "B" Clipper Circuit - The "B" clipper circuit shall sample the system "B" clipper outputs and shall provide DC voltage outputs to indicator lamps.

(7) 20 V DC Circuit - The 20 V DC circuit shall sample the plus and minus 20-volt power supply output voltages of the navigation set and shall provide DC voltage outputs to the plus and minus 20-volt indicator lamps.

3.4.3 Cable Assembly W1 of AN/APM-148 - The cable assembly shall meet the following requirements:

3.4.3.1 Functions - The cable assembly shall connect between 3J4 on Converter, Signal Data CV-971/APN-130 and 2J1 on the control-indicator. The cable shall provide the test parameter inlet from the signal data converter.

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3.4.3.2 Form Factor - The cable assembly shall be a single run cable that is 20 feet ± 2 inches in length.

3.4.3.3 Weight - The weight of the cable assembly shall not exceed 10 pounds.

3.4.3.4 Contents - The cable assembly shall consist of two connectors, necessary wires, covering, and identification bands. The two connectors shall be Bendix PT06A-20-41PY and SP06A-20-16SW or approved equivalents. The cable contains 10 double-shielded, type II (Specification MIL-W-5086), AN20 wires.

3.4.4 Cable Assembly W2 of AN/APM-148 - The cable assembly shall meet the following requirements:

3.4.4.1 Function - The cable assembly shall connect between 4J4 on Power Supply PP-2712/APN-130 and 2J2 on the control-indicator. The cable shall provide the power supply test parameter inlet to the test set.

3.4.4.2 Form Factor - The cable assembly shall be a single run cable that is 20 feet ± 2 inches in length.

3.4.4.3 Weight - The weight of the cable assembly shall not exceed 5 pounds.

3.4.4.4 Contents - The cable assembly shall consist of two connectors, necessary wires, covering and identification bands. The two connectors shall be Bendix PT06A-20-16P and SP06A-18-11S or approved equivalents. The cable contains nine wires; three 16 gauge and six 20 gauge.

3.4.5 Cable Assembly W3 of AN/APM-148 - The cable assembly shall meet the following requirements:

3.4.5.1 Function - The cable assembly shall connect between 3J3 on Converter, Signal Data CV-971/APN-130 and 2J3 on the control-indicator. The cable shall provide the test parameter inlet from the signal data converter.

3.4.5.2 Form Factor - The cable assembly shall be a single run cable that is 20 feet ± 2 inches in length.

3.4.5.3 Weight - The weight of the cable assembly shall not exceed 15 pounds.

3.4.5.4 Contents - The cable assembly shall consist of two connectors, necessary wires, covering, and identification bands. The two connectors shall be Bendix PT06A-20-41PW and SP06A-20-16S or approved equivalents. The cable contains 13 double-shielded type II (Specification MIL-W-5086), AN20 wires.

3.4.6 Cable Assembly W4 of AN/APM-148 - The cable assembly shall meet the following requirements:

3.4.6.1 Function - The cable assembly shall connect between 1J1 on the doppler signal simulator and external supply. The cable assembly shall provide primary power inlet to the test set.

3.4.6.2 Form Factor - The cable assembly shall be a single run cable that is 8 feet ± 3 inches in length.

3.4.6.3 Weight - The cable assembly shall not exceed two pounds.

3.4.6.4 Contents - The cable assembly shall consist of one connector, necessary wire, covering, and identification bands. One end shall be terminated in a pig tail. The connector shall be a MS3106E-10SL-3S or an approved equivalent. The cable contains three 16 gauge wires.

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3.4.7 Cable Assembly W5 of AN/APM-148 - The cable assembly shall meet the following requirements:

3.4.7.1 Function - The cable assembly shall connect between 2J4 on the control-indicator and 1J2 on the doppler signal simulator panel. The cable shall provide power inlet to the control-indicator.

3.4.7.2 Form Factor - The cable assembly shall be a single run cable that is 20 feet ± 2 inches in length.

3.4.7.3 Weight - The weight of the cable assembly shall not exceed six pounds.

3.4.7.4 Contents - The cable assembly shall consist of two connectors, necessary wire, covering, and identification bands. The two connectors shall be Bendix PT06A-22-21S and PT06A-22-21PW or approved equivalents. The cable contains 18 wires; four 18 gauge and fourteen 20 gauge.

3.4.8 Truck, Hand, Platform - The hand truck shall be Ryan Part Number 400377-1 or an approved equivalent and shall meet the following requirements:

3.4.8.1 Function - The hand truck shall provide for transport of the doppler radar test set during checkout of the Navigation Set, Radar AN/APN-130(V) installed in the aircraft.

3.4.8.2 Form Factor - The hand truck shall not exceed 30 inches wide by 12 inches high by 60 inches long. A layer of synthetic rubber, 1/2 inch thick, Specification MIL-R-6855, Class 2, shall be cemented to the top surface of the hand truck.

3.4.8.3 Weight - The weight of the hand truck shall not exceed 210 pounds.

3.4.8.4 Contents - The hand truck shall consist of casters and a metal frame and a layer of synthetic rubber conforming to Specification MIL-R-6855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

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4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

4.2.3.1 Rain Test - The test equipment shall perform satisfactorily after being subjected to a Rain Test in accordance with Specification MIL-E-5272 in lieu of the Watertight Tests of Specification MIL-T-21200.

4.2.3.2 Drop Test - The test equipment shall perform satisfactorily after being subjected to a Drop Test consisting of two parts as follows:

Part 1 - With the test set supported 18 inches above a concrete floor, one end shall be dropped while the other remains supported.

Part 2 - With the test set supported 18 inches above a concrete floor, release one end and drop the other end concurrently with the first impact. Repeat with ends switched.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture, shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Quality Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 CPS (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractural action)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

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4.4.3 Reliability Assurance Tests - Equipments which have successfully passed the individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094.

4.4.3.1 Test Level - The test level shall be level II from Table I of Specification MIL-R-23094.

4.4.3.2 Accept-Reject Criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.3.3 Performance Characteristics to be Measured

Characteristic	Specification Requirements	Acceptance Requirement	Time in Cycle That Measurement is to be Made
Frequency Stability	Generate Simulated Doppler return of 103.4, 990 and 1980 CPS ± 5%	± 6%	Prior to Completion of "ON" period

4.4.3.4 Preventative Maintenance - During the period of the tests the following preventative measures may be performed upon the equipment:

1. Lubricate every 200 hours of operation.
2. Replace motor brushes and bearing every 1000 hours of operation.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

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4.4.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity. The test procedure for the reliability tests shall be prepared by the contractor and approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The test equipment covered by this specification shall be for flight line use and shall provide for testing and evaluation of the Navigation Set, Radar AN/APN-130(V) while the navigation set is installed in its normal operating configuration in an aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

6.3 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

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6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

OFFICIAL BUSINESS

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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